DATA FOR AI APPLICATIONS

LLM-Assisted Environmental Data Workflows for Nature-Centred Design

25/26 MaAI01 program

Credits:Generated via NanoBanna Pro

Syllabus

This seminar introduces architecture and design students to a practical, structured approach for working with environmental and ecosystem-related data using large language models as active technical collaborators. It is intended for students without a primary background in software development who nevertheless need to formulate data-driven questions, construct lightweight analytical workflows, and critically evaluate AI-generated outputs within ecological and architectural contexts.

Responding directly to the MaAI term's focus on nature and ecosystems, the seminar positions data not as an abstract technical layer, but as a design material. It supports decision-making across environmental analysis, remote sensing, ecological performance, and climate-responsive interventions. Students will engage with environmental datasets, satellite imagery, sensor and IoT data streams, synthetic data generation, and retrieval-based AI workflows relevant to the built environment.

The course does not aim to train students as software developers in the conventional sense. Instead, it equips them with the practical and critical skills needed to work confidently with AI-assisted environmental analysis.

Students will learn how to organize a small project clearly, work with data responsibly, use code with understanding rather than imitation, and evaluate whether an AI-generated output is meaningful, misleading, or genuinely useful for design.

Across seven sessions, the seminar follows a clear progression. Students begin by defining a focused environmental or ecosystem-related design question, then identify and assess relevant data sources and their limitations. They proceed to set up a reproducible workflow, build simple AI-assisted data pipelines, and iteratively test and refine them. The course then introduces more advanced techniques, including retrieval systems, multi-model workflows, and the integration of synthetic data, culminating in the translation of results into a design-relevant output.

By the end of the seminar, each participant will have developed a small but complete AI-assisted environmental data project tied to an architectural or urban question. This includes clear documentation of methodology, assumptions, limitations, and outcomes, reinforcing a rigorous, evidence-based approach to design in the context of environmental intelligence.

Learning Objectives

At course completion, the student will be able to:

- Frame environmental or ecosystem-related design questions that can be explored through data and AI-assisted workflows.
- Identify, assess, and document relevant data sources—including environmental datasets, remote sensing inputs, sensor data, and synthetic data—while understanding their limitations.
- Structure and manage a small, reproducible technical project using appropriate tools, workflows, and documentation practices.
- Apply foundational Python skills and AI-assisted methods to build and refine simple environmental data pipelines.
- Use large language models critically as collaborators, evaluating and improving generated code, data processes, and outputs.
- Implement basic retrieval and multi-model workflows to combine data sources and analytical approaches.
- Translate technical findings into clear, design-relevant outputs for architectural and environmental applications.

Hardware / Software requirements

Participants are expected to have access to:

- A laptop capable of running Python and standard development tools.
- A GitHub account.
- Python 3.x installed.
- A code editor such as VS Code.
- Access to ChatGPT, Claude, or a comparable large language model.
- Basic spreadsheet software for quick data inspection.

Students may also work with publicly available environmental datasets, remote sensing imagery, sensor logs, or instructor-provided sample datasets to reduce setup friction and keep the focus on workflow, interpretation, and design application.

Faculty

Ahmed Al Amawi is a Principal Engineer at Chexy, a Canadian fintech company, where he leads the design and implementation of payment infrastructure, authentication systems, and data pipelines at scale. His work focuses on building reliable, production-grade systems using modern technologies including TypeScript, Python, and Google Cloud.

He has previously founded and worked on AI-focused startups and has implemented large-scale AI systems within production environments, particularly in areas such as data pipelines, automation, and AI-assisted workflows.

Alongside his industry work, he has taught and served as a teaching assistant for AI courses at IAAC in Barcelona, supporting master's-level students in developing practical and critical approaches to artificial intelligence. He is also the inventor on a published patent (WO2022183279A1).

His work sits at the intersection of software engineering, applied AI, and design, with a focus on making technical workflows more accessible without reducing rigor. In his teaching, he emphasizes clear problem framing, reproducible processes, and critical evaluation of AI outputs. Through this seminar, he guides students in building structured, testable environmental data workflows that can support design decisions.

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Basic schedule: activities and faculty

PROGRAM SCHEDULE

Tuesday April 14 · 1:00–4:00 PM (3 hours)

- **Problem Framing and Development Setup**
  - **Lecture**: Introduction to the seminar's systematic approach. Environmental data as a design material. The CRISP-DM lifecycle applied to architectural and ecological questions. What separates a vague idea ("use AI for sustainability") from a tractable data question. Live demonstration: setting up a project from scratch—Python, virtual environments, Git initialization, GitHub, and clean project folder structure.

- **Workshop**: Students form teams of 3–4. Each team sets up their full development environment and drafts a one-page problem brief identifying: the environmental question, the design decision it supports, the intended user, and measurable success criteria.
- **Deliverables**: GitHub repository with clean project structure (README, /data, /notebooks, /outputs, requirements.txt); one-page problem brief (Markdown, committed to repo); working development environment confirmed.

Tuesday, April 21 · 1:00–4:00 PM (3 hours)

- **Data Landscape and Documentation**
  - **Lecture**: The environmental data landscape: remote sensing and satellite imagery (Sentinel-2, Landsat, MODIS), climate reanalysis (ERA5), biodiversity records (GBIF), air quality (OpenAQ), IoT sensor streams, and Google Earth Engine. Introduction to Dataset Datasheets—structured documentation of provenance, contents, gaps, and misuse risks. Live demo: LLM-assisted dataset exploration, profiling, and datasheet drafting.
  - **Workshop**: Teams identify and access 1–2 candidate datasets. Using LLM assistance, they perform initial data profiling (shape, types, distributions, missing values, temporal and spatial coverage) and begin drafting a Dataset Datasheet covering source, provenance, methodology, limitations, biases, and licensing.
  - **Deliverables**: Data source inventory with access links and licensing; Dataset Datasheet draft; initial data profiling notebook. All committed to repo.

Thursday, April 30 · 1:00–4:00 PM (3 hours)

- **Python Foundations Through Environmental Data**
  - **Lecture**: Core Python taught through environmental examples: variables and types (temperature as float, station ID as string), lists and dictionaries (hourly readings, sensor metadata), loops and conditionals (flagging threshold exceedances), functions (encapsulating cleaning steps), and file I/O. Introduction to pandas and matplotlib. Live demo: prompting an LLM for code, reading the output line by line, and catching common errors.
  - **Workshop**: Teams apply Python fundamentals to their own datasets with LLM assistance. Tasks include: loading and inspecting data, cleaning and filtering, computing summary statistics, creating at least two visualizations, and writing at least one reusable function. Students must be able to explain any line of code.
  - **Deliverables**: Data exploration and cleaning notebook with at least two visualizations; updated Dataset Datasheet; at least one custom function documented. All committed to repo.

Tuesday, May 5 · 1:00–4:00 PM (3 hours)

- **Building Pipelines and Defensible Baselines**
  - **Lecture**: What is a baseline and why it matters. Data splitting strategies for environmental data (temporal autocorrelation, spatial leakage, seasonal patterns). Introduction to Model Cards: structured reporting of what a model does, how it was tested, where it fails. Live **demo**: building a complete baseline pipeline with an LLM—from raw data through cleaning, splitting, training a simple model, computing metrics, and saving results.

- **Workshop**: Teams build their own baseline pipelines. Emphasis on intentionally simple models—a measurable starting point, not a leaderboard score. Each pipeline must load data from the repo, run cleaning, split appropriately, train a baseline, compute at least two metrics, and be re-runnable from scratch. Teams begin their Model Card.
- **Deliverables**: Reproducible baseline pipeline script (must run from scratch); metrics output file; Model Card draft. All committed to repo.

Wednesday, May 13 · 1:00–4:00 PM (3 hours)

- **Synthetic Data and Model Orchestration**
  - **Lecture**: Synthetic data—why create it (sparse observations, edge cases, testing), how to create it (statistical sampling, LLM-assisted generation, physics-informed simulation), and the risks (distribution mismatch, false confidence, overfitting). Retrieval and model orchestration: combining retrieval steps with model steps, avoiding overfitting when combining models. Practical example: retrieving satellite vegetation indices from an API, combining with weather station data, and feeding both into an anomaly detector.
  - **Workshop**: Teams create a small synthetic dataset for their project covering edge cases or underrepresented scenarios, and document the generation methodology, assumptions, and validation approach. They then extend their pipeline to incorporate at least one additional data source or model component.
  - **Deliverables**: Synthetic data generation script with methodology documentation; updated pipeline with retrieval or multi-source orchestration; updated Model Card. All committed to repo.

Tuesday, May 19 · 1:00–4:00 PM (3 hours)

- **Debugging, Stress-Testing, and Failure Analysis**
  - **Lecture**: The structured debugging mindset: reading Python tracebacks, isolating whether the problem is in data, code, or model, forming and testing hypotheses, using print statements and assertions. Error taxonomy for environmental models: domain shift, sensor drift, spatial mismatch, class imbalance, temporal leakage. Live demo: the instructor deliberately breaks a working pipeline in three ways and walks through diagnosis and fix for each.
  - **Workshop**: Teams stress-test their own pipelines. Each team builds a Failure Gallery: at least 5 cases where their model or pipeline produces wrong, misleading, or unreliable results, with written diagnosis for each. Teams run at least two robustness tests (missing data, geographic shift, seasonal change, noisy input). The instructor assigns additional stress conditions for any team reporting zero failures.
  - **Deliverables**: Failure Gallery (minimum 5 cases with diagnosis); robustness test notebook with at least 2 scenarios; updated Model Card with honest limitations. All committed to repo.

Tuesday, May 26 · 1:00–4:00 PM (3 hours)

- **Decision-Facing Outputs and Final Presentations**
  - **Lecture (30 min):** Decision-facing visualization—the difference between a model output and a design tool. Communicating uncertainty honestly. Nielsen's usability heuristics as a checklist for dashboards and decision reports. What "actionable" means for different audiences. Final submission requirements overview.

- **Workshop (60 min):** Teams finalize their decision-facing output: dashboard, annotated map, anomaly report, or design recommendation tool. Each output must display intended use, non-intended use, and top three failure modes. Paired UX reviews: each team evaluates another team's output using Nielsen's heuristics.
- **Final Presentations (90 min)**: Each team presents their complete project (10 min presentation + 5 min Q&A). Presentations must cover: the environmental question, data and limitations, pipeline and baseline, what they learned from failures, and the decision-facing output.
- **Deliverables**: Decision-facing output (dashboard, report, or visualization); final presentation delivered; peer UX review completed (written feedback to partner team).

Exercise

Each team selects an environmental or ecosystem-related design question and builds a complete, LLM-assisted data workflow across all seven sessions. The project is continuous—each session's deliverables build directly on the previous session's work, and all artifacts are committed to a shared GitHub repository that serves as the living record of the project's evolution.

By Session 7, each team will have built a small but complete AI-assisted environmental data tool: from problem definition through data documentation, baseline modelling, synthetic data creation, stress testing, and a final design-facing output—all with full documentation of process, assumptions, limitations, and outcomes.

Example project directions:

- **Urban heat island mapping:** using satellite thermal imagery and weather station data to detect and visualize neighbourhood-scale heat intensity for urban planning interventions.
- V**egetation health monitoring**: tracking NDVI time series from Sentinel-2 to identify stressed urban greenery and prioritize maintenance for landscape architects.
- **Indoor air quality anomaly detection**: processing IoT sensor streams (CO_2, PM2.5, temperature, humidity) to flag ventilation failures for facility managers.
- **Biodiversity pattern analysis**: analyzing species occurrence records from GBIF to map biodiversity corridors and inform ecological connectivity design.
- **Microclimate comfort prediction:** combining weather data, site geometry, and vegetation coverage to predict outdoor thermal comfort zones for public space design.
- **Building energy anomaly detection:** analyzing hourly meter data to identify energy waste patterns and recommend retrofit priorities.

Deliverables

The final submission package is due one week after the final session (by Monday, June 2, 2026) and consists of:

- GitHub Repository containing all code, notebooks, data samples, and documentation—with a clear commit history showing the project's evolution across all seven sessions.
- Problem Brief (one page): the environmental question, the design decision it supports, the intended user, and measurable success criteria.
- Dataset Datasheet: structured documentation of all data sources, provenance, limitations, biases, and licensing.

- Model Card: structured documentation of what the model does, how it was tested, where it fails, what it should not be used for, and its honest limitations.
- Failure Gallery: minimum 5 documented failure cases with diagnosis and implications.
- Decision-Facing Output: a dashboard, annotated visualization, report, or design tool that a building or environmental professional could act on.
- README with Run Instructions: clear documentation enabling someone to clone the repo, install dependencies, and reproduce the pipeline from scratch.

Students are requested to submit all the material on the IAAC Gdrive and a Blog post need to be curated on blog.iaac.net for each project (see "IAAC | Publication Guidelines") within a maximum of 1 week after the end of the Seminar.

Grading System

• 0 - 4.9 Fail (submission of a supplementary work by January)
• 5.0 - 6.9 Pass
• 7.0 - 8.9 Good
• 9.0 - 10 Excellent/Distinction.

Evaluation, Assessment, and Final Grade

Students will be evaluated according to the following criteria:

- Attendance and participation - 15 %*
- Blog posts and Submissions 20 %*
- Academic performance - 65%
  Articulated into:
    - Demonstration of systematic, critical thinking in problem framing and data assessment—not just technical output.
    - Thorough documentation: datasheet completeness, model card honesty, failure gallery depth.
    - Reproducibility: clean repository structure, meaningful commit history, pipeline runs from scratch.
    - Quality of the decision-facing output: clarity, interpretability, uncertainty communicated, actionable for a professional.
    - Ability to explain LLM-generated code and defend analytical choices verbally.
    - Effort, timely delivery, and constructive engagement in group discussions and peer reviews.

Assessment rewards systematic thinking over model sophistication. A team that uses an LLM to build a simple model but documents it thoroughly, tests it honestly, and presents it clearly will outscore a team that prompts a complex model with no evaluation.

**Attendance and completion of submissions are mandatory, students failing to submit or not being present to at least 75% of the classes, without justification included in IAAC Terms & Conditions, will fail.*
Students who submit after the deadlines defined by the faculty and coordination will be subject to penalty and the grade will be automatically lowered by 0.3 point for every day of delay.
Incomplete submission is considered as missing submission.

References / Bibliography

- Gebru, T. et al. Datasheets for Datasets. Communications of the ACM, 64(12), 86–92. 2021.
- Mitchell, M. et al. Model Cards for Model Reporting. Proceedings of FAT* 2019.
- Wirth, R. & Hipp, J. CRISP-DM: Towards a Standard Process Model for Data Mining. 2000.
- Nielsen, J. 10 Usability Heuristics for User Interface Design. Nielsen Norman Group, 1994.
- Miller, C. et al. The Building Data Genome Project 2. Scientific Data, 7(1), 2020.
- Gorelick, N. et al. Google Earth Engine: Planetary-Scale Geospatial Analysis. Remote Sensing of Environment, 202, 2017.
- Naboni, E. & Havinga, L. Regenerative Design in the Digital Practice: A Handbook for the Built Environment. 2019.
- Further readings and dataset documentation provided in each session.